

$$= + 3.1 \times 10^{-5} \text{ J}$$

18. Ans: A

When the electron moves downward, its gravitational PE decreases. (The Earth's gravitational field is pointing vertically downward.)

The electron has negative charge, so it will accelerate towards the positive plate due to the downward electric force on it. Its gain in KE comes mostly from its loss of electric PE. So its electric PE decreases.

(H2 students can also reason this way: when a negative charge moves from a region of lower electric potential to another region of higher electric potential, its electric PE decreases.)

19. Ans: B